

Recognise that complementary events have a combined probability of one; use this relationship to calculate probabilities in applied contexts

Learning intention

- recognise that complementary events have a combined probability of one; use this relationship to calculate probabilities in applied contexts.

Teach and model

- using the sum of probabilities to solve problems.
- using the relationship that for a single event A , $\Pr(A) + \Pr(\text{not } A) = 1$.
- understanding that knowing the probability of an event allows the probability of its complement to be found.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.